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BASIC OF SIGNAL PROSESSING (BEC402)

**ESP32-Based Audio Signal Visualizer
Using MAX4466**

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ABSTRACT

Signal processing plays an important role in modern electronic and communication systems. It involves the acquisition, analysis, manipulation, and interpretation of signals such as audio, video, temperature, and other physical parameters. In embedded systems, signal processing techniques are widely used for real-time monitoring and automation applications.

This project, “ESP32-Based Audio Signal Visualizer Using MAX4466”, demonstrates a simple implementation of real-time audio signal processing using an ESP32 microcontroller and a MAX4466 microphone module. The microphone captures surrounding sound signals and converts them into analog electrical signals. These signals are then processed by the ESP32 using its inbuilt Analog-to-Digital Converter (ADC).

Depending on the amplitude of the detected sound, LEDs glow in different patterns to visually represent sound intensity. Lower sound levels activate fewer LEDs, while louder sounds activate more LEDs. Thus, the system acts as an audio level indicator or sound visualizer.

The project demonstrates important concepts of Basic Signal Processing such as:

- signal acquisition,
- analog signal processing,
- amplitude detection,
- ADC conversion,
- input-output response,
- and real-time embedded processing.

The system is low-cost, compact, and suitable for educational demonstrations and mini-project applications.

1. INTRODUCTION

This project presents the design and implementation of a real-time audio signal visualizer using an ESP32 microcontroller, MAX4466 microphone module, and LEDs. The MAX4466 microphone captures surrounding sound signals and converts them into analog voltage signals. These analog signals are continuously sampled by the ESP32 ADC.

The ESP32 processes the incoming audio signal and compares the detected signal amplitude with predefined threshold values. Based on the intensity of the sound, one or more LEDs glow to visually indicate the audio level. The system therefore converts audio signals into visible light patterns.

The project demonstrates fundamental concepts of Basic Signal Processing including signal acquisition, analog-to-digital conversion, amplitude analysis, and real-time system response. The implementation provides a simple and cost-effective method for understanding embedded signal processing applications.

The developed system can be extended further for applications such as:

- music visualizers,
- sound-reactive lighting,
- noise monitoring systems,
- and embedded audio analysis systems.

2. PROBLEM STATEMENT

In modern electronic and embedded systems, real-time signal monitoring and analysis are essential for various applications such as audio processing, noise monitoring, smart automation, and sound-controlled devices. Traditional audio signal processing systems are often expensive, complex, and require advanced hardware and software resources.

There is a growing need for a simple, low-cost, and real-time embedded system capable of:

- acquiring audio signals from the environment,
- processing analog sound data,
- analyzing sound intensity,
- and visually representing audio levels in an understandable form.

However, implementing such a system involves several challenges:

Challenges Faced

- Capturing continuously varying sound signals accurately.
- Converting analog audio signals into digital form for processing.
- Performing real-time signal analysis using embedded hardware.
- Generating quick visual feedback corresponding to sound intensity.

- Designing a compact and low-cost system suitable for educational purposes.

To address these issues, an ESP32-based audio signal visualization system using a MAX4466 microphone module and LEDs is proposed. The system processes environmental sound signals and visually displays sound intensity using LEDs in real time.

3. OBJECTIVES

1. Audio Signal Acquisition

- To capture surrounding sound signals using the MAX4466 microphone sensor.
 - To convert acoustic signals into analog electrical signals.
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2. Analog Signal Processing

- To process continuously varying analog sound signals using the ESP32 microcontroller.
 - To understand real-time signal variation and amplitude changes.
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3. ADC Implementation

- To utilize the inbuilt ADC (Analog-to-Digital Converter) of ESP32.
 - To convert analog microphone signals into digital values for processing.
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4. Sound Intensity Detection

- To detect different sound levels based on signal amplitude.
 - To compare incoming signal values with predefined threshold levels.
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5. Visual Representation of Signals

- To display sound intensity using LEDs.
 - To provide real-time visualization of audio amplitude.
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6. Implementation of BSP Concepts

- To demonstrate practical applications of:
 - signal acquisition,
 - signal analysis,
 - amplitude detection,
 - real-time embedded processing,
 - and system response. .

4. PROPOSED SYSTEM

The proposed system is a real-time audio signal visualization system developed using an ESP32 microcontroller, a MAX4466 microphone module, and LEDs. The main purpose of the system is to capture surrounding sound signals, process them using embedded signal processing techniques, and visually indicate sound intensity through LEDs.

The system is designed to be:

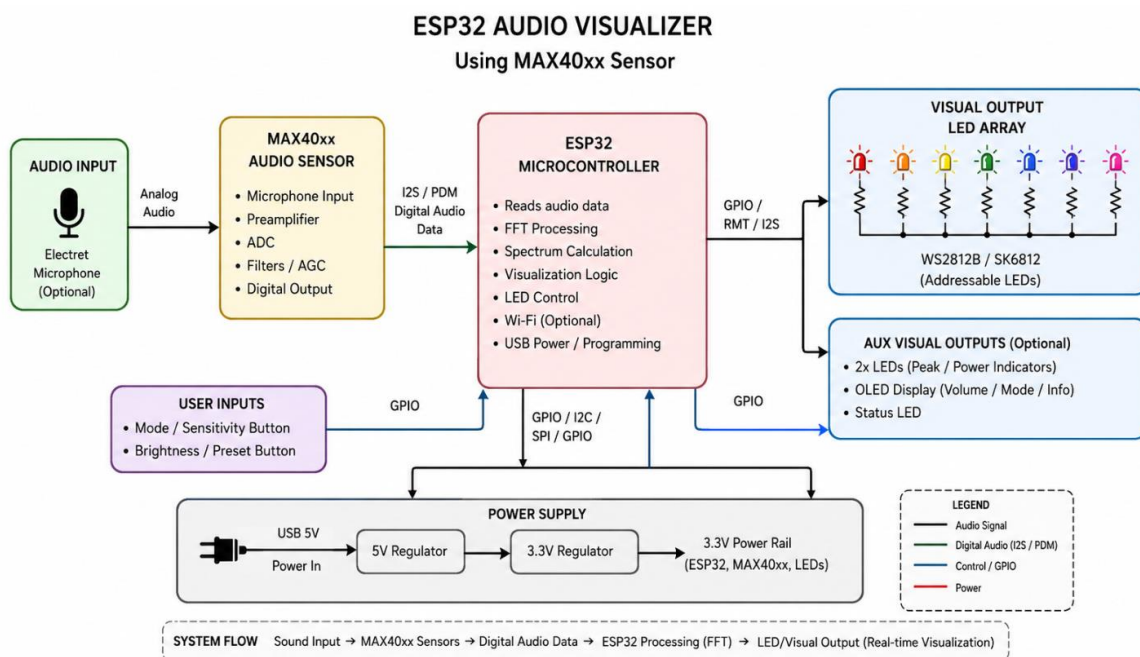
- low-cost,
- compact,
- easy to implement,
- and suitable for educational applications related to Basic Signal Processing (BSP).

5. SYSTEM ARCHITECTURE

The system architecture of the proposed project describes the overall organization and interaction between different hardware and software components used in the audio signal visualization system.

The system mainly consists of:

- Input Unit,
- Processing Unit,
- Signal Conversion Unit,
- and Output Unit.



The architecture explains how the sound signal flows from the microphone to the LED output through the ESP32 microcontroller.

Page | 8 **1.Input Unit**

MAX4466 Microphone Module

The MAX4466 microphone module acts as the input device of the system.

Functions:

- Captures environmental sound signals.
- Converts sound waves into analog electrical signals.
- Sends varying analog voltage signals to the ESP32.

Input Signals:

- Speech
- Music
- Claps
- Environmental noise

The microphone continuously generates analog signals proportional to sound intensity.

2. Signal Conversion Unit

ESP32 ADC (Analog-to-Digital Converter)

The ESP32 contains an inbuilt 12-bit ADC.

Functions:

- Receives analog signals from the microphone.
- Converts analog voltages into digital values.

- Generates ADC readings between:

0 to 4095

Purpose:

This enables the ESP32 to process real-world analog sound signals digitally.

3. Processing Unit

ESP32 Microcontroller

The ESP32 acts as the main processing unit of the system.

Functions:

- Reads ADC values continuously.
- Processes incoming sound signals.
- Compares sound amplitude with threshold values.
- Controls LED outputs according to sound intensity.

Processing Operations:

- Signal acquisition
- ADC sampling
- Amplitude analysis
- Threshold comparison
- Output control

The ESP32 executes the embedded C program uploaded through Arduino IDE.

4. Output Unit

LED Visualization System

Three LEDs are used as visual indicators of sound intensity.

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Working:

- Low sound → 1 LED ON
- Medium sound → 2 LEDs ON
- High sound → 3 LEDs ON

The LEDs provide real-time visualization of the detected audio signal amplitude.

HARDWARE ARCHITECTURE

Hardware Component Purpose

MAX4466	Audio signal acquisition
ESP32 ADC	Analog-to-digital conversion
ESP32 Controller	Signal processing
LEDs	Visual output
Resistors	Current limiting

SOFTWARE ARCHITECTURE

The software part of the system consists of:

- initialization routines,
- ADC reading functions,
- threshold comparison logic,

- and LED control instructions.

Main Software Functions

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Function	Purpose
analogRead()	Reads microphone signal
digitalWrite()	Controls LEDs
if conditions	Threshold comparison
delay()	Controls processing speed

OVERALL SYSTEM OPERATION

1. Environmental sound is captured by the MAX4466 microphone.
2. The microphone converts sound into analog voltage signals.
3. ESP32 ADC converts analog signals into digital values.
4. ESP32 processes the signal in real time.
5. Signal amplitude is compared with threshold values.
6. LEDs glow according to detected sound intensity.

FEATURES OF THE SYSTEM ARCHITECTURE

- Real-time signal acquisition
- Embedded signal processing
- ADC-based audio analysis
- Simple hardware design
- Low-cost implementation

- Real-time visual output
- Expandable architecture for advanced DSP applications

6. HARDWARE COMPONENTS

1. ESP32 Development Board

The ESP32 acts as the main controller of the system. It reads analog sound signals from the microphone using ADC, processes the signal, and controls LEDs according to sound intensity.

2. MAX4466 Microphone Module

The MAX4466 microphone module captures environmental sound such as speech, music, and noise. It converts sound waves into analog voltage signals for processing.

3. LEDs

The LEDs provide visual representation of audio intensity. More LEDs glow as sound intensity increases.

4. 220Ω Resistors

These resistors protect LEDs by limiting the amount of current flowing through them.

5. Breadboard

The breadboard is used to build the circuit without soldering and helps in easy testing and modifications.

6. Jumper Wires

Jumper wires establish electrical connections between all hardware components.

7. USB Cable

The USB cable is used for:

- powering the ESP32,
- uploading the Arduino code,
- and serial communication.

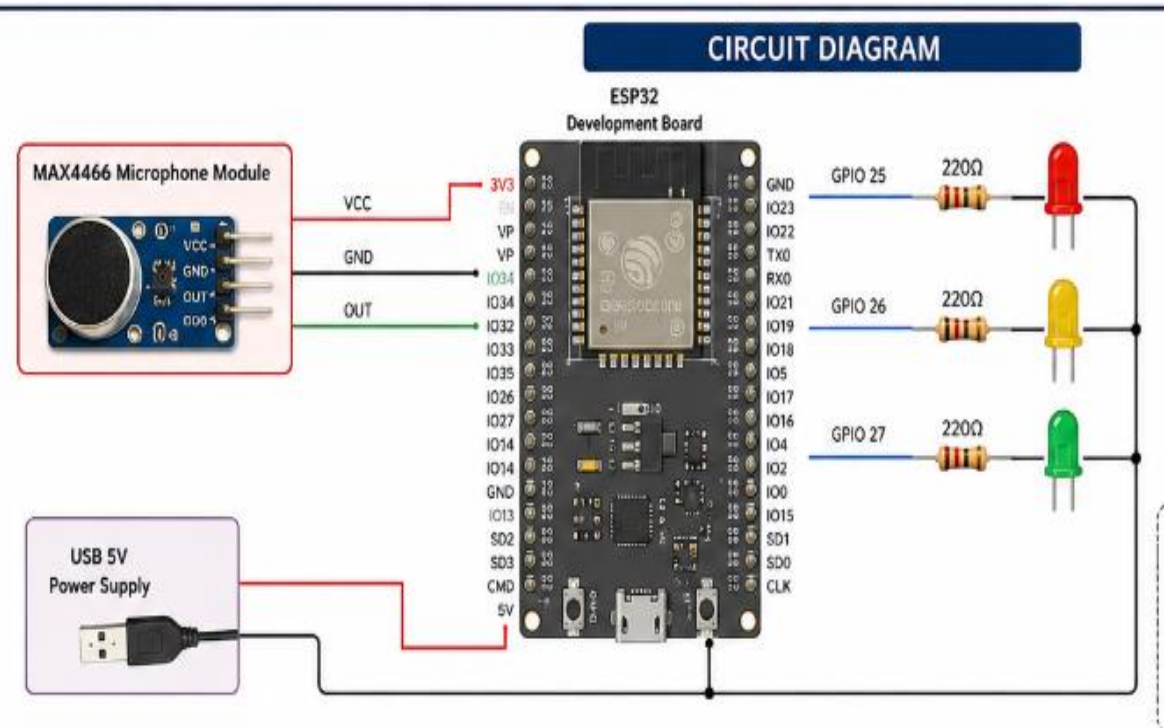
8. Power Supply

The entire system operates using USB 5V power supplied to the ESP32. The ESP32 internally provides 3.3V to the microphone module and GPIO pins.

HARDWARE COMPONENTS LIST

Sl. No.	Hardware Component	Quantity	Purpose / Use	Image
1	ESP32 Development Board	1	Main microcontroller used for signal processing, ADC conversion, and LED control	
2	MAX4466 Microphone Module	1	Captures environmental sound signals and converts them into analog electrical signals	
3	LEDs	3	Used for visual indication of sound intensity levels	
4	220Ω Resistors	3	Limits current flow to protect LEDs from excess current	
5	Breadboard	1	Used for temporary circuit assembly and component connections	
6	Jumper Wires	Multiple	Used for connecting ESP32, microphone module, resistors, and LEDs	
7	USB Cable	1	Provides power supply and program uploading to ESP32	
8	Power Supply (USB 5V)	1	Supplies electrical power to the ESP32 and connected components	

7. CIRCUIT DIGRAM AND ITS METHODOLOGY



The circuit diagram of the ESP32-based Audio Visualizer consists of three major sections: the input section, processing section, and output section. The system is designed to capture environmental sound signals using a microphone module, process the sound data using the ESP32 microcontroller, and visually indicate sound intensity through LEDs. The complete circuit works in real time and demonstrates basic signal processing concepts such as signal acquisition, analog-to-digital conversion, amplitude detection, and output visualization.

The input section of the circuit consists of the MAX4466 microphone module. This module is responsible for capturing surrounding sound signals such as speech, music, claps, and

environmental noise. The microphone converts these sound waves into analog electrical signals. The MAX4466 module contains an inbuilt amplifier which amplifies weak audio signals for better processing. The module has three main connections: VCC, GND, and OUT. The VCC pin is connected to the 3.3V pin of the ESP32, the GND pin is connected to the common ground, and the OUT pin is connected to the GPIO39 (VN) analog input pin of the ESP32.

The processing section consists of the ESP32 microcontroller, which acts as the brain of the entire system. The ESP32 continuously receives analog audio signals from the microphone module. Since the microphone produces analog voltage signals, the ESP32 uses its inbuilt Analog-to-Digital Converter (ADC) to convert these analog signals into digital values. The ADC of the ESP32 provides digital readings in the range of 0 to 4095. These digital values represent the amplitude or intensity of the incoming sound signal. Higher sound levels generate larger ADC values, while lower sound levels generate smaller values.

After receiving the digital signal, the ESP32 processes the data using the embedded program uploaded through the Arduino IDE. The program continuously reads the ADC values using the `analogRead()` function. These values are then compared with predefined threshold levels to determine the sound intensity. The threshold values help classify the sound into different categories such as low sound, medium sound, and high sound.

The output section of the circuit consists of three LEDs connected to the ESP32 GPIO pins through 220 Ω current-limiting resistors. LED1 is connected to GPIO18, LED2 is connected to GPIO19, and LED3 is connected to GPIO21. The negative terminals of all LEDs are connected to the ground. The

resistors are used to protect the LEDs from excess current and ensure safe operation.

The LEDs act as visual indicators of sound intensity. When the sound level is low, only the first LED glows. When the sound intensity increases, the second LED also glows. For very high sound levels such as loud music or claps, all three LEDs glow simultaneously. In this way, the LEDs provide a real-time visualization of audio amplitude.

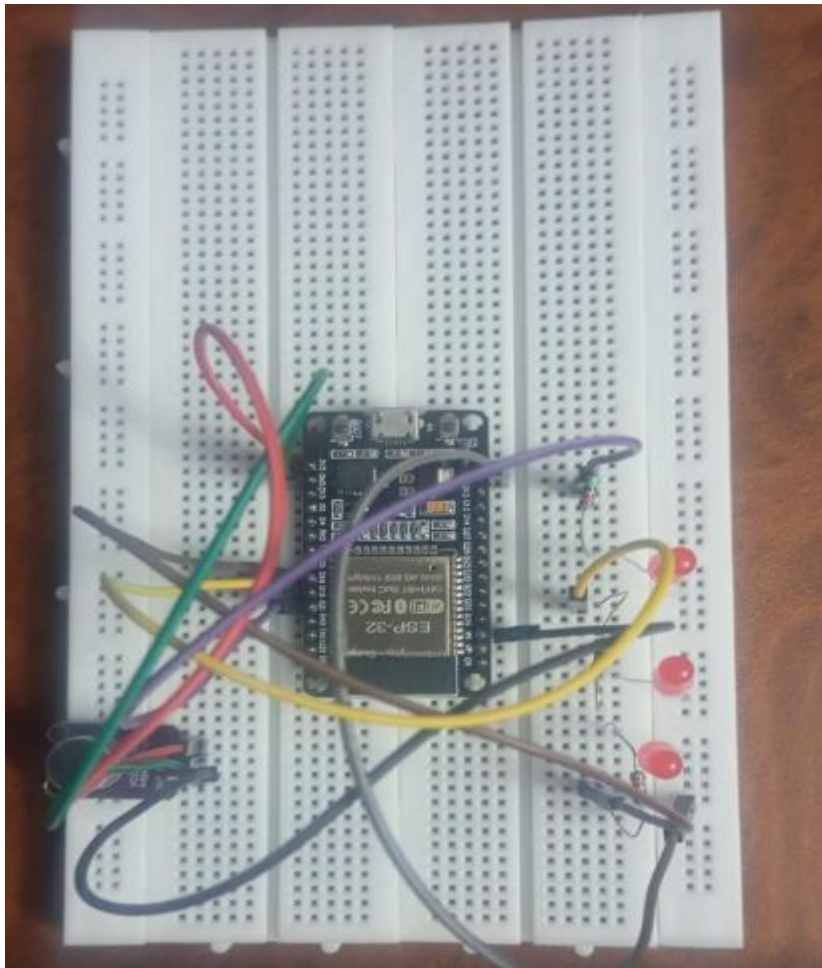
The entire system is powered using a USB cable connected to the ESP32. The ESP32 receives 5V power through the USB connection and internally regulates it to provide 3.3V supply to the microphone module and GPIO pins. The breadboard and jumper wires are used to establish temporary electrical connections between all components.

The methodology of the project begins with sound acquisition. The MAX4466 microphone continuously captures environmental sound and converts it into analog electrical signals. These analog signals are transmitted to the ESP32 ADC input pin. The ESP32 continuously samples the incoming signal and converts it into digital values using ADC. The microcontroller then processes the sound data and compares the signal amplitude with predefined threshold values.

Based on the detected sound intensity, the ESP32 controls the LEDs accordingly. Lower sound intensity activates one LED, medium sound intensity activates two LEDs, and higher sound intensity activates all three LEDs. This process continues repeatedly in real time, creating an audio visualization effect.

Overall, the circuit demonstrates the practical implementation of embedded signal processing techniques using low-cost hardware components. The project successfully combines audio

signal acquisition, ADC conversion, signal analysis, and visual output generation into a compact real-time embedded system.



8.WORKING PRINCIPLE

1. Audio Signal Acquisition

The MAX4466 microphone module acts as the input sensor for the system.

Function:

- Captures environmental sound waves.
- Converts acoustic signals into analog electrical signals.

The microphone detects:

- speech,
- claps,
- music,
- fan noise,
- and surrounding environmental sounds.

The output of the microphone is a continuously varying analog voltage signal proportional to sound intensity.

2. Analog Signal Transmission

The analog output signal generated by the MAX4466 module is connected to the analog input pin of the ESP32.

Connection:

MAX4466 OUT → ESP32 GPIO39 (VN)

The microphone continuously sends varying analog voltage levels to the ESP32 ADC input.

3. Analog-to-Digital Conversion (ADC)

The ESP32 contains an inbuilt 12-bit ADC (Analog-to-Digital Converter).

ADC Function:

- Converts analog voltages into digital values.
- Produces digital readings between:

0 to 4095

Interpretation:

ADC Value	Sound Level
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Small value	Low sound
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Medium value	Medium sound
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Large value	High sound
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The ADC continuously samples incoming sound signals in real time.

4. Signal Processing

The ESP32 processes the digital sound values using embedded programming logic.

Process:

- The ESP32 continuously reads ADC values using:
analogRead()
- The obtained values are analyzed in real time.
- The program compares the signal amplitude with predefined threshold levels.

Example:

```
if(sensorValue > threshold)
```

This allows the system to classify sound intensity into different levels.

5. Threshold-Based Detection

Different sound intensity levels are represented using threshold values.

Example Thresholds

Sound Intensity Threshold

Low sound 2200

Medium sound 2500

High sound 2800

When the incoming sound signal exceeds a threshold, the corresponding LED is activated.

6. LED Visualization

Three LEDs are used to visually indicate the detected sound intensity.

LED Operation

Sound Level	LED Response
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Low sound	LED 1 ON
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Medium sound	LED 1 & LED 2 ON
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High sound	All LEDs ON
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Thus, the LEDs act as a real-time visual representation of audio signal amplitude.

7. Real-Time Operation

The entire system operates continuously in real time.

Real-Time Features

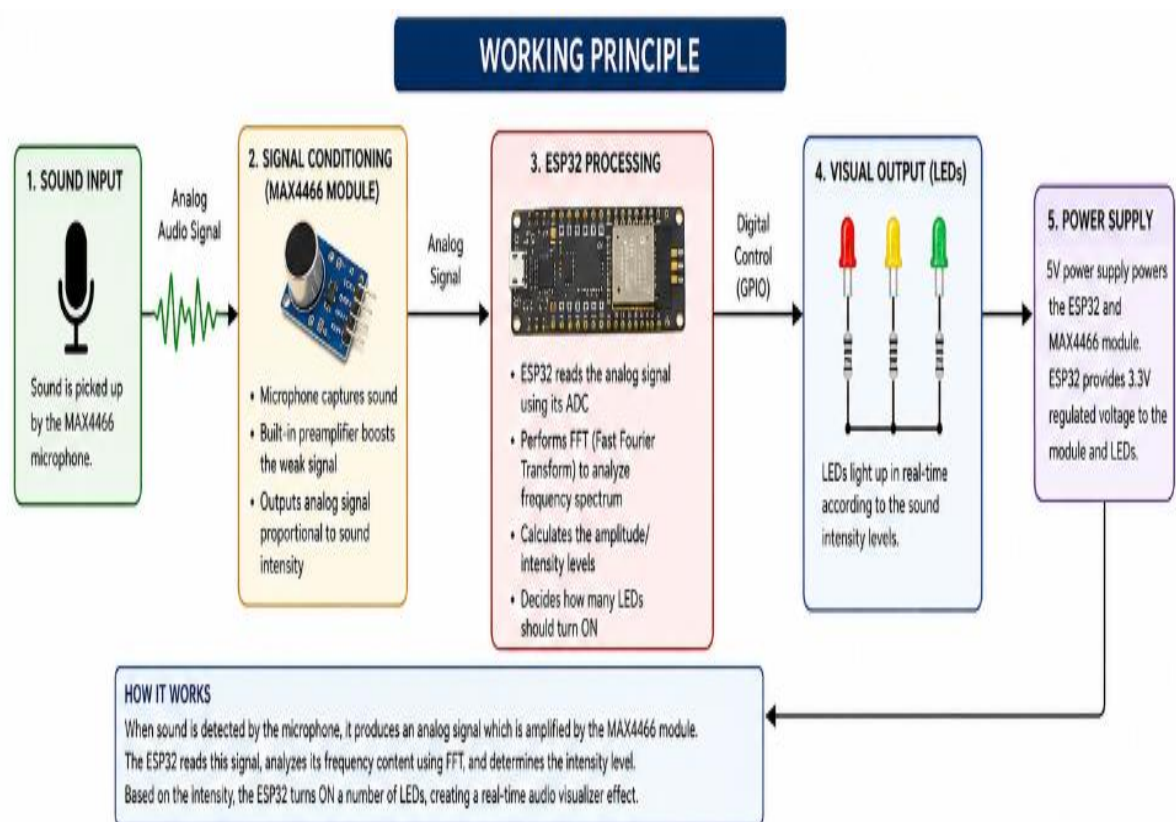
- Continuous microphone sampling
- Instant signal processing
- Immediate LED response

- Dynamic audio visualization

The ESP32 repeatedly:

1. reads audio signals,
2. processes ADC values,
3. compares thresholds,
4. controls LED outputs.

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9. USE OF BASIC SIGNAL PROCESSING'S METHODOLOGIES

1. Signal Acquisition

- The MAX4466 microphone module captures environmental sound signals.
- Sound waves are converted into analog electrical signals.
- Demonstrates real-time signal acquisition.

2. Analog Signal Processing

- The microphone generates continuously varying analog signals.
- ESP32 processes these analog audio signals in real time.
- Demonstrates handling of time-varying signals.

3. Analog-to-Digital Conversion (ADC)

- ESP32 uses its inbuilt 12-bit ADC.
- Converts analog microphone signals into digital values.
- ADC output range:
0 to 4095
- Demonstrates practical ADC implementation.

4. Signal Amplitude Analysis

- ESP32 continuously reads sound amplitude values.
- Higher sound intensity produces larger ADC values.
- Demonstrates amplitude detection and signal analysis.

5. Threshold-Based Signal Detection

- ADC values are compared with predefined threshold values.
- Different sound levels are classified based on amplitude.
- Demonstrates signal comparison and decision-making.

6. Real-Time Signal Processing

- ESP32 continuously samples and processes sound signals.
- LEDs respond instantly to changing sound intensity.
- Demonstrates real-time embedded processing.

7. Input-Output System Response

- Microphone acts as input signal source.
- ESP32 acts as processing system.
- LEDs act as output response system.
- Demonstrates input-output behavior of a signal processing system.

8. Dynamic System Behavior

- Output changes continuously with varying sound levels.
- Demonstrates characteristics of a dynamic system.

9. Time-Varying System

- Sound signals continuously change with time.
- LED output also changes accordingly.
- Demonstrates time-varying signal processing.

10. Causal System

- Output depends only on present sound input.
- No future signal values are required.
- Demonstrates causal system operation.

11. Embedded Signal Visualization

- Sound amplitude is visually represented using LEDs.
- Demonstrates practical signal visualization techniques.

10. APPLICATIONS

1. Audio Level Indicator System

The project can be used as a simple audio level indicator where LEDs display the intensity of surrounding sound. It can be used in classrooms, studios, and small audio systems to monitor sound levels visually.

2. Music Visualization Systems

The system can be used as a basic music visualizer where LEDs glow according to music beats and sound intensity. It can be integrated into speakers, DJ systems, and decorative lighting applications.

3. Noise Monitoring Systems

The project can be used for monitoring environmental noise levels in:

- classrooms,
- offices,
- libraries,

- hospitals,
- and industrial areas.

The LEDs can provide instant indication of high noise levels.

4. Sound-Controlled Automation

The system can be extended for sound-based automation applications such as:

- clap-controlled lights,
- sound-activated appliances,
- smart home systems,
- and voice-triggered devices.

5. Embedded Signal Processing Demonstration

The project serves as an educational demonstration for:

- analog signal processing,
- ADC conversion,
- amplitude analysis,
- and real-time embedded systems.

It is highly suitable for laboratory experiments and academic mini-projects.

6. Smart Lighting Systems

The audio visualization system can be integrated into decorative lighting systems where LEDs respond dynamically to surrounding sound or music.

11. ADVANTAGES

ADVANTAGES OF ESP32 AUDIO VISUALIZER

ADVANTAGE	DESCRIPTION
 Low-Cost Implementation	Uses inexpensive and easily available electronic components.
 Real-Time Operation	Processes and displays sound intensity instantly.
 Simple Circuit Design	Easy to assemble and understand for beginners.
 Compact and Portable	Small-sized system that can be easily carried and used anywhere.
 Easy Programming	Program can be developed easily using Arduino IDE.
 Low Power Consumption	Consumes less electrical power during operation.
 Educational Value	Demonstrates practical Basic Signal Processing concepts.
 Expandable System	Can be upgraded with IoT, FFT, and advanced visualization features.
 Fast Response	LEDs respond quickly to changing sound levels.
 Practical Implementation	Provides hands-on experience in embedded systems and signal processing.

12. FUTURE SCOPE

- ☐ Implementation of FFT-based frequency analysis
- ☐ Wireless monitoring using Wi-Fi and IoT
- ☐ Mobile application integration
- ☐ Smart home automation systems
- ☐ Advanced RGB LED visualization

- ☐ Noise pollution monitoring systems
- ☐ Voice recognition and speech processing
- ☐ Cloud-based sound data monitoring
- ☐ AI and machine learning-based audio analysis
- ☐ Advanced DSP application development

13. CONCLUSION

The ESP32-based Audio Signal Visualizer using the MAX4466 microphone module successfully demonstrates the implementation of real-time audio signal processing and visualization. The system captures environmental sound signals, converts them into digital values using the ESP32 ADC, and visually represents sound intensity using LEDs. The project effectively implements important Basic Signal Processing concepts such as signal acquisition, analog-to-digital conversion, amplitude analysis, and real-time system response. The developed system is simple, economical, easy to implement, and suitable for educational as well as embedded system applications..

14. REFERENCES

- ☐ ESP32 Datasheet – Espressif Systems
- ☐ MAX4466 Microphone Amplifier Module Datasheet
- ☐ Arduino IDE Documentation
- ☐ Basic Signal Processing Notes and Laboratory Manual

□ ESP32 ADC Documentation

□ Arduino Official Website

□ Embedded Systems and Signal Processing Reference 28□

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Online Tutorials and Technical Articles on ESP32 and MAX4466
Interfacing